

Product Requirements Document (PRD)

Product Name

[NeuronDash]

Product Vision

Build an AI-powered data analytics platform that allows students, businesses, researchers, and professionals to upload Excel, CSV, PDF, and Word files and automatically transform them into clean datasets, interactive dashboards, AI-generated reports, and actionable insights.

The platform should eliminate the need for manual data cleaning, complex spreadsheet analysis, and traditional business intelligence tools.

Problem Statement

Users often have valuable data stored inside spreadsheets, PDFs, and documents but lack the technical expertise to:

- Clean messy datasets
- Identify patterns and trends
- Create professional dashboards
- Generate reports
- Extract meaningful insights

Current solutions require extensive manual work, technical skills, or expensive business intelligence software.

The platform will solve this by using AI to automate the entire workflow.

Target Users

Primary Users

Students

- Academic projects
- Research work
- Assignment analysis
- Performance tracking

Businesses

- Sales analysis
- Revenue tracking
- Operational reporting
- Performance monitoring

Researchers

- Survey analysis
- Data interpretation
- Research reporting

HR Teams

- Employee analytics
- Attendance tracking
- Performance evaluation

Finance Teams

- Financial reporting
- Budget analysis
- Profitability tracking

Core Features

1. File Upload

Users can upload:

- Excel (.xlsx)
- CSV (.csv)
- PDF (.pdf)
- Word (.docx)

Capabilities:

- Drag and drop upload
 - Multi-file upload
 - File validation
 - Upload progress tracking
-

2. AI Data Cleaning

After upload, AI automatically analyzes datasets and identifies:

- Missing values
- Duplicate records
- Invalid formats
- Inconsistent entries
- Empty columns
- Outliers

Cleaning actions:

- Remove duplicates
- Fill missing values
- Standardize formats
- Normalize categories
- Correct data inconsistencies

Users can review and approve cleaning suggestions before processing.

3. AI Chat Interface

The upload experience will function as a conversational AI workspace.

Users can interact with uploaded data through prompts.

Examples:

- Analyze this dataset
- Find anomalies
- Create an executive dashboard
- Generate a performance report
- Show key trends
- Summarize insights

The AI responds with recommendations and generated outputs.

4. Dashboard Generation

Users can generate dashboards automatically from uploaded datasets.

Supported dashboard templates:

Executive Dashboard

- KPI cards
- Growth metrics
- Trend analysis
- Summary insights

Sales Dashboard

- Revenue tracking
- Product performance
- Regional analysis
- Sales funnel

Student Dashboard

- Attendance analysis
- Academic performance
- Subject comparison
- Grade distribution

Finance Dashboard

- Revenue
- Expenses
- Profitability
- Cash flow

HR Dashboard

- Employee metrics
- Attendance
- Attrition
- Department performance

Research Dashboard

- Survey analytics
- Response distribution
- Demographic analysis
- Trend visualization

AI Auto Dashboard

AI automatically selects the most appropriate dashboard based on uploaded data.

5. AI Insight Generation

The platform automatically generates:

- Key findings
- Trend analysis
- Performance indicators
- Anomaly detection
- Business recommendations
- Data summaries

Insights should be displayed alongside dashboards.

6. AI Report Generation

Generate professional reports from uploaded data.

Report sections:

Executive Summary

Key Findings

Trends

Opportunities

Risks

Recommendations

Reports should be editable before export.

7. Export & Sharing

Users can export:

- PDF Reports
- Dashboard Snapshots
- Cleaned Data Files

Sharing options:

- Public Link
 - Team Sharing
 - Download
-

User Flow

Landing Page



Sign Up / Login



Dashboard Home



Upload File + AI Chat



Data Preview



Data Cleaning Review



Dashboard Template Selection



Dashboard Generation



AI Insight Generation



AI Report Generation



Export / Share

Dashboard Home Requirements

Display:

- Recent uploads
- Generated dashboards
- Generated reports
- Quick actions
- Usage statistics

Quick actions:

- Upload New File
 - Generate Dashboard
 - Create Report
-

Functional Requirements

Authentication

- User Registration
- User Login
- Password Reset
- Secure Sessions

File Processing

- Upload validation
- File storage
- Parsing engine
- Data extraction

AI Processing

- Dataset understanding
- Cleaning suggestions
- Dashboard recommendation
- Report generation
- Conversational querying

Dashboard Engine

- Template selection
 - Auto-generation
 - Interactive charts
 - Filters
 - Responsive design
-

Non-Functional Requirements

Performance

- File upload under 10 seconds
- Dashboard generation under 30 seconds
- Report generation under 20 seconds

Scalability

- Support thousands of users
- Support large datasets

Security

- Encrypted file storage
- Secure authentication

- Role-based access
- Data privacy compliance

Reliability

- 99.9% uptime target
 - Automatic backups
-

Future Roadmap

Phase 2 Features:

- Chat with Data
- Scheduled Reports
- Team Collaboration
- Dashboard Sharing
- AI Forecasting
- PowerPoint Export
- API Integrations

Phase 3 Features:

- Multi-Workspace Support
 - Enterprise Permissions
 - White Label Dashboards
 - Predictive Analytics
 - Industry-Specific Templates
-

Success Metrics

- File Upload Completion Rate
- Dashboard Generation Rate
- Report Export Rate
- User Retention
- Monthly Active Users
- Subscription Conversion Rate

MVP Scope

Launch Version Includes:

- ✓ Authentication
- ✓ File Upload
- ✓ AI Data Cleaning
- ✓ AI Chat Interface
- ✓ Dashboard Generation
- ✓ 7 Dashboard Templates
- ✓ AI Insights
- ✓ AI Report Generation
- ✓ Export Functionality
- ✓ Responsive Design

Everything else belongs in future releases.